University
Mohammed VI
Polytechnic

College of
Computing

Deliverable #: Lab 6: SQL – DDL and DML

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

Session: Fall 2026

Team Information

Team Name	The Avengers
Member 1	Valdez Kemekong
Member 2	Ranya El Bahali
Member 3	Hayat Ennani
Member 4	Aouatif Guennoun
Repository Link	`https://github.com/aouatif-Gu/database`

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1 Introduction

In this lab we are supposed to implement the final relational schema RLMS using SQL Data Definition Language (DDL), also we need to populate and modify the RLMS database using SQL Data Manipulation Language (DML) retrieve information using SELECT queries, which are part of the DML, progressing from simple single-table queries to joins and grouped aggregations. In addition to that, we need to deliver a short report that describes the role of each team member, summarizes the main implementation difficulties encountered, explains any assumptions made while implementing or populating the database, and answers the open question of Part 4.

2 Requirements

Task 1 : Create SQL tables using the final RLMS relational schema obtained in the previous labs.
Task 2 : Populate the RLMS database with enough data to meaningfully test the queries in Task 3 and then perform some DML operations.
Task 3 : Write and execute SQL queries that are ordered approximately from simpler single-table queries to joins and grouped aggregations.
Task 4 : Answer the open question.

3 Methodology

MEMBER ROLES :
HAYAT ENNANI : Worked on part 1 with the rest of the other members, wrote the code for the whole task 2 part and gave her assumptions and feedback about what she learned in the process.
AOUATIF GUENNOUN : Worked on the part A B of the task number 3 , wrote the reports after combining the assumptions and challenged faced from all the other memebers and made the latex file.
RANYA EL BAHALI : worked on the pat C and D of the task number 3 and helped the other members do the first task by giving her ideas and assumptions.
VALDEZ KEMEKONG : worked on the E F parts in task number 3 and the open question in task number 4 , and helped the other members figuring out the solution for task 1.

ASSUMPTIONS :
for part C COALESCE(..., 0): Converts NULL result into 0 justification without coalesce the consumables that were never been used return null due to the left join applying sum to missing rows resukts in null rather than a zero.

4 Implementation Results

THE CODE IS PROVIDED IN SEPERATE FILES IN THE GITHUB REPO
Part 4 :Open question

1-Budgets and funds

The total sum of funds allocated (FundsPrj and FundsLab) for a given budgetline must not exceed the total amountGranted in the BudgetTable. FRom what we have seen so far, SQL cannot enforce such cross-table constraints during insertion,which requires the database has to use something like a trigger to validate the sum in real time.

2-reservations and equipments

An equipment must not be reserved by two reservations at the same time. From what we have seen so far, SQL can't check existing bookings during insertion

3- supervisor-supervisee

An employee can't be his own supervisor. When writing the id of the supervisor as a foreign key in an instance of employer table, it only proves that the supervisor is a true employer but it doesn't check if it's actually the employer itself. From what we have seen so far, it's impossible to prevent that.

5 Discussion

CHALLENGES:
RANYA: the main challenges that i faced is the order of execution was a bit confusing and choosing the right join type For instance, if a laboratory has zero equipment units, using an inner join instead of a left join can accidentally drop that laboratory from the aggregation output
HAYAT: The only thing I was confused about is when to use ON DELETE NO ACTION CASCADE.
AOUATIF : I found it hard to manipulate the joins and to predict what would be the solution for each query.
VALDEZ: Expressing the division at the F was quite difficult so I used another method which is counting the number of certifications needed for each equipment that need a certification and find the persons whose number of certifications for a model was actually the number of certifications needed for that model.

6 Conclusion

In this lab we have implemented the final RLMS relational schema using SQL Data Definition Language (DDL), populated and modify the RLMS database using SQL Data Manipulation Language (DML). and rtrieved information using SELECT queries, which are part of the DML, progressing from simple single-table queries to joins and grouped aggregations.